Geosynthetics

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
Woven Geotextile	IS 16391 (Part 1); IS 16391;IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 10318-1 (Geosynthetics — Terms and Definitions — Geotextiles and Geotextile-related Products); ISO 10319 (Geosynthetics — Wide-width Tensile Test)	ASTM D4595 (Tensile Properties of Geotextiles by the Wide-Width Strip Method); ASTM D4491 (Water Permeability of Geotextiles); ASTM D4751 (Apparent Opening Size of Geotextiles); ASTM D6241 (Static Puncture Strength of Geotextiles and Geogrids)	DIN EN ISO 10319 (Geotextiles — Wide-Width Tensile Test); DIN EN ISO 12956 (Determination of Characteristic Opening Size); DIN EN ISO 11058 (Water Permeability Characteristics); DIN EN ISO 12236 (CBR Puncture Test)	GRI GT Series (Geosynthetic Research Institute Specifications); FHWA Geosynthetics Guidelines (Highway Applications); AASHTO M288 (Geotextile Specification for Highway Applications)
Nonwoven Geotextile	IS 16391 (Part 2); IS 16391;IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 10318-1 (Geosynthetics — Terms and Definitions — Geotextiles and Geotextile-related Products); ISO 10319 (Geosynthetics — Wide-width Tensile Test)	ASTM D4632 (Grab Breaking Load and Elongation of Geotextiles); ASTM D4491 (Water Permeability of Geotextiles); ASTM D4751 (Apparent Opening Size of Geotextiles); ASTM D6241 (Static Puncture Strength of Geotextiles and Geogrids)	DIN EN ISO 10319 (Wide-Width Tensile Test); DIN EN ISO 12956 (Characteristic Opening Size); DIN EN ISO 11058 (Water Permeability); DIN EN ISO 12236 (CBR Puncture Resistance)	GRI GT Series; FHWA Geosynthetic Guidelines; AASHTO M288
Geogrid (Uniaxial); Geogrid (Biaxial)	IS 17373 (Geogrids for Civil Engineering Applications); IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 10319 (Geosynthetics — Wide-width Tensile Test); ISO 12956 (Geotextiles and Geotextile-related Products — Determination of Characteristic Opening Size)	ASTM D6637 (Determining Tensile Properties of Geogrids by Single or Multi-Rib Tensile Method); ASTM D7737 (Standard Test Method for Individual Geogrid Junction Strength); ASTM D5262 (Evaluating Unconfined Tension Creep Behavior of Geosynthetics)	DIN EN ISO 10319 (Wide-Width Tensile Test); DIN EN ISO 13431 (Geotextiles and Geotextile-Related Products — Tensile Creep Properties); DIN EN ISO 12960 (Resistance to Alkaline, Acid and Saline Liquids, where applicable); DIN EN ISO 12224 (Tensile Creep Behaviour, where applicable)	GRI GG Series (Geogrid Specifications); FHWA Geosynthetics Guidelines; AASHTO R Standards for Geosynthetics
Geonet	IS 17372 (Geosynthetics for Drainage Applications); IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 12958-1 (Geotextiles and Geotextile-related Products — Determination of Water Flow Capacity — In-plane Flow Under Constant Head); ISO 10319 (Wide-width Tensile Test)	ASTM D4716 (Determining the In-Plane Flow Rate per Unit Width and Hydraulic Transmissivity of Geosynthetic Drainage Products); ASTM D5199 (Measuring the Nominal Thickness of Geosynthetics); ASTM D7005 (Determining the Bond Strength of Geosynthetic Clay Liners)	DIN EN ISO 12958 (Geotextiles — Determination of In-Plane Water Flow Capacity); DIN EN ISO 10319 (Tensile Properties); DIN EN ISO 13438 (Screening Test for Oxidation Resistance); DIN EN ISO 12236 (CBR Puncture Resistance)	GRI GN Series (Geonet Specifications); EPA Subtitle D (Landfill Applications); FHWA Geosynthetics Guidelines
Geocell	IS 17371 (Geocells for Soil Reinforcement Applications); IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 10319 (Geosynthetics — Wide-width Tensile Test); ISO 12957-1 (Geosynthetics — Determination of Friction Characteristics — Direct Shear Test)	ASTM D638 (Tensile Properties of Plastics); ASTM D1505 (Density of Plastics); ASTM D5199 (Thickness Measurement of Geosynthetics); ASTM D5261 (Measuring Mass per Unit Area of Geotextiles)	DIN EN ISO 10319 (Wide-Width Tensile Test); DIN EN ISO 12236 (CBR Puncture Resistance); DIN EN ISO 13431 (Tensile Creep Properties); DIN EN ISO 527-1 (Tensile Properties of Plastics, where polymer sheet testing applies)	GRI GC Series (Geocell Specifications); FHWA Geosynthetics Guidelines; AASHTO Geosynthetic Reinforcement Guidelines
Geocomposite Drain; Drainage Composite	IS 17372 (Geosynthetics for Drainage Applications); IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	;ISO 12958-1 (Geotextiles and Geotextile-related Products — Water Flow Capacity — In-plane Flow Under Constant Head); ISO 10318-1 (Geosynthetics — Terms and Definitions)	ASTM D4716 (Hydraulic Transmissivity of Geosynthetic Drainage Products); ASTM D5199 (Thickness of Geosynthetics); ASTM D5261 (Mass per Unit Area of Geotextiles); ASTM D1621 (Compressive Properties, where core materials apply)	DIN EN ISO 12958 (In-Plane Water Flow Capacity); DIN EN ISO 11058 (Water Permeability Characteristics); DIN EN ISO 10319 (Tensile Properties); DIN EN ISO 12236 (Puncture Resistance)	GRI GC Series; FHWA Geosynthetic Drainage Guidelines; EPA Subtitle D (Landfill Drainage Systems)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
HDPE Geomembrane	IS 16352;IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 9862 (Geosynthetics — Sampling and Preparation of Test Specimens); ISO 527-3 (Plastics — Film and Sheeting — Tensile Properties)	ASTM D5199 (Measuring Nominal Thickness of Geosynthetics); ASTM D6693 (Tensile Properties of Plastic Geomembranes); ASTM D5397 (Evaluation of Stress Crack Resistance of Polyolefin Geomembranes); ASTM D5885 (Oxidative Induction Time of Polyolefin Geosynthetics)	DIN EN ISO 527-3 (Tensile Properties of Plastic Films); DIN EN ISO 1183-1 (Density of Plastics); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 13438 (Oxidation Resistance)	GRI GM13 (HDPE Geomembrane Specification); EPA Subtitle D (Waste Containment Applications); NSF/ANSI 61 (Water Containment, where applicable)
LLDPE Geomembrane	IS 16352;IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 9862 (Geosynthetics — Sampling and Preparation of Test Specimens); ISO 527-3 (Plastics — Film and Sheeting — Tensile Properties)	ASTM D6693 (Tensile Properties of Plastic Geomembranes); ASTM D5199 (Thickness of Geosynthetics); ASTM D1004 (Initial Tear Resistance of Plastic Film and Sheeting); ASTM D5397 (Stress Crack Resistance of Polyolefin Geomembranes)	DIN EN ISO 527-3 (Tensile Properties of Plastic Films); DIN EN ISO 1183-1 (Density); DIN EN ISO 527-1 (Tensile Properties); DIN EN ISO 13438 (Oxidation Resistance)	GRI GM17 (LLDPE Geomembrane Specification); EPA Subtitle D; FHWA Geosynthetics Guidelines
Geosynthetic Clay Liner	IS 17370 (Geosynthetic Clay Liners); IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 10769 (Geosynthetics — Test Method for Measuring Mass Per Unit Area of Geosynthetic Clay Liners); ISO 10318-1 (Geosynthetics — Terms and Definitions)	ASTM D5887 (Measurement of Index Flux Through Saturated Geosynthetic Clay Liner Specimens); ASTM D5891 (Fluid Loss of a Geosynthetic Clay Liner); ASTM D5993 (Measuring Mass per Unit Area of Geosynthetic Clay Liners)	DIN EN ISO 9863-1 (Geosynthetics — Determination of Thickness); DIN EN ISO 10772 (Filtration Behaviour); DIN EN ISO 13426-1 (Internal Structural Integrity of Geosynthetic Clay Liners); DIN EN ISO 12958 (In-Plane Water Flow Capacity)	GRI GCL Specifications; EPA Subtitle D; FHWA Geosynthetics Guidelines
Prefabricated Vertical Drain	IS 15284 (Prefabricated Vertical Drains); IS 16352; IS 14151; IS 16017; IS 16738; IS 15351; IS 15909; IS 16361; IS 16428; IS 16490	ISO 18325 (Geosynthetics — Test Method for Measuring the Hydraulic Conductivity of Prefabricated Vertical Drains); ISO 12958-1 (Geotextiles and Geotextile-related Products — Determination of Water Flow Capacity)	ASTM D4716 (Hydraulic Transmissivity of Geosynthetic Drainage Products); ASTM D1621 (Compressive Properties of Rigid Cellular Plastics, where applicable); ASTM D5261 (Mass per Unit Area of Geotextiles); ASTM D6916 (Determining Interface Shear Strength of Geosynthetic Materials, where applicable)	DIN EN ISO 12958 (In-Plane Water Flow Capacity); DIN EN ISO 10319 (Tensile Properties); DIN EN ISO 9863-1 (Thickness); DIN EN ISO 13431 (Tensile Creep Properties)	FHWA Geosynthetics Guidelines; GRI Standard Practice for Drainage Composites; AASHTO Geotechnical Standards

Source field: Plastic and Plastic Products